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Principal Component Analysis



Recap: Understanding PCA Fundamentals

A Quick Look at Principal Component Analysis (PCA)

What is PCA?

- Unsupervised technique to reduce data dimensions
- Finds directions of highest variance
- Projects data into fewer dimensions with minimal information loss

Why Reduce Dimensions?

- Tackle the “curse of dimensionality”
- Lower computational load
- Remove multicollinearity
- Make complex data easier to visualise

Core Concepts

- **Variance:** Spread of data, PCA aims to maximise this
- **Eigenvectors:** Directions of maximum variance
- **Eigenvalues:** Variance in each direction
- **Orthogonality:** Components are at right angles, reducing overlap

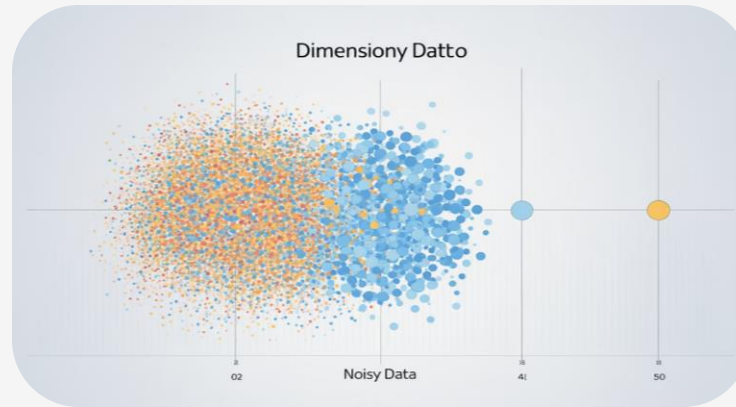
Real-World Applications of PCA

How Principal Component Analysis Makes Data Work Smarter



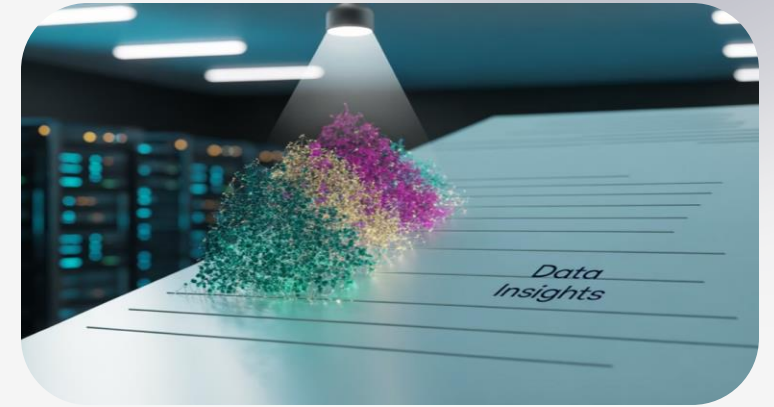
Image Compression

- Uses fewer dimensions to represent images
- Speeds up facial recognition by removing redundant details
- Boosts processing and model performance



Noise Reduction

- Keeps only high-variance features
- Filters out low-variance noise
- Enhances clarity in signals and genomics



Data Visualisation

- Converts complex data into 2D or 3D views
- Reveals patterns, clusters, and outliers
- Makes high-dimensional data easier to explore

PCA reduces feature space, helping models run faster, avoid overfitting, and generalise better

PCA: Challenges and Limitations

What to Watch out for When Using Principal Component Analysis

Loss of Interpretability

- Principal components are linear combinations of the original features, which can make them difficult to interpret
- “Black box” effect is a problem when understanding each feature is critical
- Example: In healthcare, a component may mix blood pressure, cholesterol, and BMI in ways that are unclear to doctors

Limited to Linear Relationships

- PCA assumes linear patterns between features and may miss more complex, non-linear relationships
- It performs poorly on curved or spiral data structures
- Example: PCA can't find an optimal low-dimensional view for S-shaped data

PCA: Challenges and Limitations

What to Watch out for When Using Principal Component Analysis

Scaling Sensitivity

- PCA is sensitive to feature scale
- Large-scale values dominate the analysis if data isn't standardised, leading to misleading outcomes
- Example: Income (in thousands) may outweigh age (in decades) in variance calculations

PCA doesn't handle categorical data without encoding, and choosing the right number of components needs domain knowledge and testing.

Step-by-Step Example – PCA in Action

MNIST Digits Dataset Example

1. Data Preprocessing

```
# Load and standardize datafrom sklearn.datasets import load_digitsfrom sklearn.preprocessing import StandardScalerdigits = load_digits()X = digits.data # 8x8 pixel images flattened to 64 dimensionsy = digits.targetX_scaled = StandardScaler().fit_transform(X)
```

2. Applying PCA

```
# Fit PCAfrom sklearn.decomposition import PCApca = PCA()X_pca = pca.fit_transform(X_scaled)
```


Beyond PCA: Dimensionality Reduction Options

Compare PCA, t-SNE and UMAP for Different Data Needs

1

PCA (Linear)

- Best for: Linear data, Gaussian distribution, and speed
- Keeps: Global structure and max variance
- Limitation: Misses non-linear patterns

2

t-SNE (Non-linear)

- Best for: Visualising clusters in 2D/3D
- Keeps: Local relationships and groupings
- Limitation: Slow, can't handle large data or new inputs

3

UMAP (Non-linear)

- Best for: Visualising clusters in 2D/3D
- Keeps: Local relationships and groupings
- Limitation: Slow, can't handle large data or new inputs



Choose a method based on your goal—visualisation, speed, or structure. Try a few and compare outcomes.

Test different techniques on the same dataset to see which reveals the most useful patterns

PCA + Supervised Learning

Problem Statement

Predicting Tumour Type Using Diagnostic Data

- Predict if a tumour is malignant or benign
- **Aim:** Create an accurate, easy-to-understand model for early detection
- **Challenge:** Too many features can cause overfitting and poor clarity
- **Solution:** Apply PCA to reduce features, then use Logistic Regression for classification

Dataset Description

Breast Cancer Wisconsin (Diagnostic) Dataset

- **Source:** UCI Machine Learning Repository
- **Samples:** 569 breast cancer cases
- **Features:** 30 numerical values per sample
 - Based on digitised fine needle aspirate (FNA) images
 - Include metrics like radius, texture, perimeter, area and smoothness
- **Target:**
 - 1 = Malignant (cancerous)
 - 0 = Benign (non-cancerous)

Why Use PCA Here?

Benefits Before Classification

- Reduces noise and keeps the most useful feature combinations
- Prevents overfitting from correlated or redundant features
- Makes visualisation and interpretation easier
- Speeds up training and boosts model generalisation

Q&A Session



Open floor for questions and clarifications.